

ToxinAudit-LabelScope: an enzyme EC provenance gate before family-transfer modeling

Candidate-specific working report, September 25, 2026. This report does not extend the previous toxin family's AUROC result or count that earlier work as a new result. It records a new exact-accession enzyme evidence audit. Its principal scientific outcome is that a predictive benchmark should not be run without measured function-negative labels, even after the positive accession set was expanded.

Abstract

A random-versus-family generalization gap in the earlier VenomFunction project prompted a distinct question: would a new protein-function benchmark with enzyme EC labels also reveal annotation shortcuts when sequence families are held out? We inspected direct UniProtKB reviewed entries for carboxylesterase EC 3.1.1.1 and hexokinase EC 2.7.1.1. Twenty-six individual entry JSON records were fetched, with original URLs and SHA-256 hashes. Seven esterase and five hexokinase entries have an experimental ECO:0000269 citation attached specifically to the target EC. Fourteen other entries have the target EC annotation but no direct experimental evidence attached to that specific target EC in the parsed entry. None is a measured function-negative control. A subsequent frozen API-ranked expansion independently fetched 120 reviewed EC entries, 62 esterases and 58 hexokinases, with 22 and 19 target-EC experimental citations respectively. The original 26 are preserved but not added to the 120. Neither panel contains assayed target-function negatives. A 3-mer sequence-overlap proxy finds 22 clusters in the original panel and at least one cluster crossing a seeded random train/test split, but no classifier or AUROC can be justified from a panel with no assayed negatives. Direct QuickGO, InterPro and ChEMBL crosschecks provide annotation context, not negative function measurements. The output is a runnable fail-closed benchmark auditor, not a validated enzyme-function model.

1. Biological question and distinction from the parent toxin work

The earlier VenomFunction analysis compared random splits to snake-toxin family-disjoint splits using a curated keyword. A large gap there is motivation for a different annotation task, not an independent test of this enzyme problem. Enzyme Commission (EC) terms describe reactions under specified substrates and conditions; they cannot be treated as measured binary labels for arbitrary proteins without examining evidence. Our two target EC values, 3.1.1.1 (carboxylesterase) and 2.7.1.1 (hexokinase), have different chemistry and enzyme-family composition. The planned question was whether a predictor trained on one set of enzyme families would maintain performance on held-out groups compared with a random split. A defensible binary benchmark requires positives with direct experimental support and negatives assayed for absence of the *same target reaction*. A protein with a different EC is not automatically negative for a multifunctional enzyme, and a record lacking ECO:0000269 is not a negative.

We used a build-time feasibility gate: at least five direct experimentally supported positives in each function class, at least five experimentally verified negatives for that function, and homolog-disjoint train/test groups before computing a predictive metric. The threshold was scoped during the build, not registered with a third party. It is a refusal criterion, not a claim that five labels are enough for precise field-scale AUROC.

2. Direct accession retrieval and label parsing

`build_data.py` requests each selected accession's UniProtKB JSON record separately at `https://rest.uniprot.org/uniprotkb/<accession>.json`. Each record in `accessions.json` stores accession, sequence, organism, taxonomy ID, all parsed EC terms, target-EC experimental evidence codes, URL and response digest. Both recommended and alternative protein names are parsed, because a target EC can appear under an alternative name. The first convenience panel was 12 reviewed proteins. Fourteen more carboxylesterase records were retrieved as a new fixed tranche, yielding 26 entries: 19 esterase and seven hexokinase. This selection is nonrandom and not representative of all enzymes.

For each record, the code checks that the expected target EC occurs among the parsed EC annotations. It then distinguishes a direct ECO:0000269 citation on *that EC* from a citation on another EC or a general annotation elsewhere in the protein. Among 19 esterase entries, seven have direct target-EC experimental citation; among seven hexokinases, five do. The positive count minimum is therefore met after expansion. The remaining 14 records must

be marked 'no direct experimental target-EC evidence in this parsed annotation', not 'experimentally inactive'. A function assay may exist outside the entry; missing proof in this source cannot prove biological absence.

The organism span includes human, rodent, plant, fly and Mycobacterium for the esterase panel. This makes a broad annotation audit possible but introduces enormous changes in protein architecture, assay substrates and sequence composition. We did not pool these organisms into a measured, harmonized function screen. Response hashes guard against silently changing remote records when rerunning the code; they do not adjudicate whether original papers' assays were comparable.

3. Approximate homology contamination audit

`audit.py` computes each sequence's set of amino-acid 3-mers and pairwise Jaccard overlap. At a fixed 0.55 overlap cutoff it takes connected components as an **approximate contamination proxy**. This is not sequence alignment, orthology assignment or a family phylogeny. It yields 22 groups for 26 entries. A deterministic random quarter-test split (seed 27) shares at least one proxy cluster across train and test. That exposes a concrete failure mode for a future random fold: a homolog-like group can occupy both sides. It does not prove the old toxin model's full AUROC difference was caused by this exact mechanism, and no new enzyme AUROC was fitted.

The threshold and split were specified as audit defaults, not chosen for predictive performance. Even if a group-disjoint assignment were made, there would still be no assayed negative class. Recording a family split alone would not rescue the benchmark. A larger panel with independent structure or curated orthology groups and harmonized enzyme assays would be needed before comparing random and family-held-out metrics.

4. Evidence crosschecks from distinct sources

To test whether two named accessions mapped to the expected annotation families, `crosschecks.py` queried [EBI QuickGO](#), [InterPro](#) and [ChEMBL](#), for O00748 and P19367. The source ledger stores response hashes and interpreted records. O00748 maps to InterPro carboxylesterase type B domain IPR002018 and ChEMBL target ChEMBL3180. P19367 maps to InterPro hexokinase family IPR001312 and ChEMBL target ChEMBL2688. QuickGO supplies molecular-function annotation evidence identifiers; its GO:0004340 records for P19367 include computational or inferred references as well as curated context. GO and domain hits are not direct absence assays. ChEMBL target indexing is not a standardized EC activity outcome for every construct.

The original scientific source inventory was UniProtKB, QuickGO, InterPro and ChEMBL (four); a later UniRef50 sequence-cluster audit adds a fifth materially used resource. EBI hosts two distinct annotated data resources; if the program counts all EBI endpoints as one conservative parent service, the count would be lower. Repeated accessions, model libraries, transport infrastructure and alternate endpoint paths do not turn five sources into forty independent tools.

5. Results and refusal gate

Evidence unit	Count or verdict	Interpretation
Direct reviewed UniProtKB entries	26	Exact accession JSON retrieved and hashed
Esterase target-EC direct experimental citations	7/19	Positive feasibility minimum reached, not a balanced cohort
Hexokinase target-EC direct experimental citations	5/7	Positive feasibility minimum reached
Experimental target-function negatives	0 documented	Essential gate missing for both function classes
Approximate 3-mer overlap components	22	Proxy grouping only, not curated families
Seeded random split crossing proxy groups	At least one group	Warns that random-split evaluation would leak similarity
Predictor AUROC	Not estimated	Deliberate fail-closed refusal

There is no performance figure to compare with the earlier VenomFunction result. A fabricated negative class from the 14 evidence-unspecified records might produce a number but would answer an annotation-provenance question, not biochemical function. The tool returns an explicit null for function-prediction AUROC and marks family holdout unassessable until label scope is repaired.

6. Implementation and tests

`build_data.py` retrieves entry JSON in parallel with bounded workers. `audit.py` checks uniqueness, target EC presence, positive evidence grade, proxy sequence groups and deterministic split crossings. `test_build_data.py` asserts that all 26 accession URLs and hashes are present and no fetched entry is silently relabeled as an experimental negative. `test_audit.py` tests duplicate refusal, wrong-target EC rejection, group formation and null AUROC. `crosschecks.py` can be rerun to refresh the three contextual annotations for two named accessions. Run `python3 projects/toxin_audit/build_data.py`, `python3 projects/toxin_audit/audit.py`, and `python3 -m pytest -q projects/toxin_audit` from repository root. Remote UniProt records may change; compare response hashes before treating a rerun as identical data.

7. Limits and next valid experiment

First, ECO:0000269 on an EC annotation is evidence-linked curation, not a complete re-analysis of substrate, rate and assay context. Second, the accession panel mixes organisms and substrates, so a numerical split comparison would confound lineage and label provenance. Third, 3-mer Jaccard components can miss homologs with rearrangements and collapse unrelated low-complexity proteins. Fourth, the convenience accession list is not sampled from a defined background population. Fifth, no cross-protein target-function absence assay was found in this original accession retrieval, and no outside wet-lab cohort has been validated. A later single-protein variant assay is scoped separately below. The correct present state is a tested source/label auditor and a blocked predictive benchmark.

A future experiment could gather independently measured positive and negative substrate assays for the same target reaction, freeze family groups from a credible alignment/orthology resource, reserve a truly external measured cohort and only then fit a fixed linear baseline. If those data are unavailable, preserving the negative feasibility result is scientifically preferable to relabeling a curated database as ground truth. This report does not promise that future data will produce a positive classifier.

Source notes

[UniProtKB REST](#) owns the direct protein entries; each individual URL is stored in `accessions.json`. [QuickGO](#), [InterPro](#) and [ChEMBL](#) supplied interpreted context in `external_checks.json`. Exact output objects are `accessions.json`, `results.json` and `external_checks.json`. No source here supplies a measured negative enzyme-function cohort.

Post-report accession expansion, September 25

The original 26-entry cohort and its numerical cluster results above remain an archival small-panel analysis. For the distinct retrieval-count gate, `expand_panel.py` froze the first 62 reviewed UniProtKB EC 3.1.1.1 and 58 reviewed EC 2.7.1.1 identifiers in search order before inspecting evidence. It then fetched 120 exact entry JSON responses separately, saving URL, response-byte SHA-256, sequence and parsed EC evidence in `data/expanded_accessions.json`; the fixed search URLs and chosen IDs are in `data/expanded_selection.json`. Included multifunctional proteins required parsing EC terms under the entry's `includes` components, not only its main name. The expanded audit finds 22/62 and 19/58 target-EC direct ECO:0000269 citations, 79 records without that grade, and 97 approximate 3-mer components. These are nonrandom, heavily cross-species accession annotations. No activity-negative assay cohort appeared; a missing citation remains unknown, so the function-prediction AUROC is null. `data/expanded_results.json` records that refusal. This establishes 120 separately retrieved accession records, not 120 independent activity assays, biological discoveries, or a benchmark win.

External sequence-cluster audit

Each of the 120 expanded enzyme accessions was separately mapped to a [UniRef50 sequence cluster](#), saving URL, response SHA-256 and ID in `data/uniref50_membership.json`. The panel spans 73 UniRef50 groups; 19 contain multiple selected entries, with a largest group of 11. The two EC target classes do not co-occur within a selected group in this mapping. This is an external 50%-identity grouping reference rather than the report's original 3-mer proxy, and a fifth materially used annotation source. It still does not establish substrate-specific negative activity or a measured family-disjoint benchmark; the AUROC remains null.

Separate measured variant-assay scope check, September 26

[Gersing et al. 2023](#) tested full-length human glucokinase GCK variants by expressing them in yeast lacking endogenous hexokinases, then selecting growth on glucose. The [original workbook](#) provides 9,362 unfloored continuous amino-acid-level scores. The paper reports coverage of 9,003/9,280 possible missense and nonsense variants and a 38-variant comparison with in-vitro catalytic efficiency (Spearman 0.76, its own result). Our independent source parsing found 8,570 missense, 433 nonsense and 359 synonymous variant records, all with HGVS reference residues matching the separately fetched 465-aa UniProt P35557 (EC 2.7.1.1) sequence. At a descriptive, **not validated classifier**, threshold of score <0.2, 2,788 missense, 332 nonsense and nine synonymous records fall below it. The original assay, score normalization, abundance effects and error distribution matter, so this is not proof that each low-score variant is biochemically inactive. One protein's thousands of related variants are one experimental dataset, not thousands of independent enzyme families, and a within-P35557 diagnostic cannot supply measured inactive *other-protein* controls for the planned family-disjoint esterase or hexokinase classifier. We make no AUROC claim and retain the no-benchmark verdict. The exact source links, workbook and sequence digests, independent parser and focused tests are in `data/gck_variant_scope.json` and `gck_variant_scope.py`.